

254 ENGINEERING SERVICES

Manufactured Home Foundation Certification Protocol

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Prepared by	Aman Dhakal, P.E., Engineer of Record, 254 Engineering Services
Approval	Aman Dhakal, P.E., Engineer of Record
Applies to	All manufactured home foundation certification engagements performed by 254 Engineering Services
Supersedes	Version 1.0, dated October 4, 2026.

1. Purpose

This protocol establishes how 254 Engineering Services inspects the foundation of an existing manufactured home so that the engineer can issue a sealed certification that the foundation complies with a named published requirement, supported by an evidence package collected by a field technician under the engineer's supervision.

The engineer signs and seals the certification personally and a lender relies on it to make a loan. Every requirement in this protocol exists so that the engineer can reach a determination from the evidence package alone.

2. Scope

This protocol applies to existing manufactured homes built to the HUD Code on or after June 15, 1976, installed on a foundation in Texas, where a lender or loan program requires an engineer's certification of the foundation.

The certification addresses the foundation, the anchorage, and the connection of the home to the foundation, and any addition or alteration as it bears on them.

This protocol does not cover the home's own structure or its compliance with the HUD Code; the installer's licensing or installation report under Texas rules; the home's title or real property status; retrofit design, which is a separate service line; or site-built and modular homes, which are covered by the foundation certification protocol. The firm does not perform forensic or cause-of-loss investigation, geotechnical investigation, or subsurface soil testing under this or any protocol.

3. Governing requirements

- The certification is written against HUD's Permanent Foundations Guide for Manufactured Housing, HUD-7584, September 1996 (PFGMH). This is the firm's default on every job. Another requirement is used only where the lender names it in writing, it is recorded verbatim at intake, and the engineer approves it before dispatch.
- FHA Single Family Housing Policy Handbook 4000.1 requires a certification by an engineer or architect licensed in the state where the home is located, attesting to compliance with the

PFGMH. The certification is site specific and carries the engineer's signature, seal, and license number.

- Where the appraiser notes an addition or alteration to the home, Handbook 4000.1 requires the certification to address it.
- A lender may reuse an earlier certification where there has been no alteration or observable damage since. That is the lender's decision. The firm does not re-date or re-address a certification.
- Where the lender names a requirement other than the PFGMH, the engineer decides before dispatch whether this protocol's evidence can support it.
- The certification is the engineer's professional opinion based on a visual, non-invasive inspection. It is not a warranty or a representation about concealed or below-grade conditions.

4. Roles and responsibilities

Only the Engineer of Record determines the outcome of a job and the wording of the certification. The field technician and the coordinator relay the engineer's written determination without interpretation, addition, or estimate of when a certification will issue.

5. Personnel qualifications

Field Technician

- At least two years of documented experience in manufactured home installation, foundation inspection, or residential construction, or equivalent experience the engineer accepts in writing.
- Completion of the engineer's training on this protocol and Appendix B, with a supervised inspection before independent work.
- Confined-space awareness training, because every job requires going under the home.
- Annual refresher training when the governing requirement, this protocol, or Appendix B changes.

Engineer of Record

- Active Texas P.E. license. A copy, with expiration date, is held in the firm's compliance file.

6. Intake

- The coordinator opens a job file for each property. The file records the intake answers and customer uploads required by Appendix A.
- The coordinator records who requires the certification and their requirement exactly as given. The certification is addressed only to that recipient and written only against that requirement.
- A yes to any flag question in Appendix A routes the job to the engineer before dispatch. The engineer records accept, accept with conditions, or decline.
- A job is not dispatched until intake is complete and every upload required by Appendix A has been received.
- The engineer records any condition of acceptance in the job file, including any item the technician must capture beyond Appendix B.

7. Scheduling and dispatch

- The coordinator schedules an accepted job within two business days. The engineer may set a different interval for a job in the job file.

- The technician confirms access under the home, occupancy, and access to the interior data plate before travel.
- Where site conditions prevent complete capture, including weather, access, safety, or standing water under the home, the technician records each item that could not be observed and the reason. No item is estimated, assumed, or left blank.
- Where the space under the home cannot be safely entered, the technician does not enter it. The engineer decides from the evidence collected whether a determination is possible.

8. Field inspection

The technician completes every item of Appendix B for the property, in the order listed, and follows the photo procedure printed at the head of that checklist.

- Before starting, the technician makes a rough sketch of the home and any additions, marks North, and photographs the sketch.
- The technician ticks the item being photographed, photographs the checklist, then takes the photographs for that item, and moves to the next item the same way.
- An item that does not apply to the property is marked with the reason it does not apply.
- The counts and measurements called for on the checklist are recorded as counts and measurements, not as descriptions.
- The technician goes under the home on every job. Piers, footings, and anchorage are recorded along the full length of each chassis beam, not from the access opening.

9. Evidence standard and engineer review

Evidence package

Each job produces one evidence package in the firm's field application, consisting of the completed Appendix B checklist, the photographs, and the technician's notes.

- Every checklist item has at least one wide photograph placing the item on the home and one close photograph showing its condition.
- Where a size, spacing, height, or clearance governs, a tape is in frame.
- The worst pier, the worst footing, and the worst strap or anchor on each side of the home are photographed. A representative sample selected for appearance is not acceptable.
- The HUD label on each section and the data plate are photographed so that every line can be read.
- Photographs are captured directly in the firm's field application and carry device time, server time, the difference between them, and location. Photographs sent by text or messaging app, or taken on another device, are not accepted.
- An observation that cannot be made is recorded as not observed with the reason.
- A package that does not cover every checklist item, with the photographs and measurements each item calls for, is rejected without review.

Engineer review

- The engineer reviews each evidence package and records one of the five determinations in Appendix C: pass, revise, repairs required, site revisit, or decline.
- Where foundation plans were supplied, the engineer compares the foundation as observed against them and records any difference.
- The engineer's review record identifies the checklist items and photographs relied on for the determination, and states the basis for accepting the anchorage.

- A package that does not allow the engineer to verify a required item is returned for revision or a site revisit.

10. Deficiencies and repairs

- On a repairs required determination, the engineer issues a written repair list identifying each item, what governs it, and what is required to close it.
- Where closing an item needs an engineered retrofit, the repair list says so. The retrofit design is a separate engagement.
- Repairs are performed by others. 254 Engineering Services does not perform, supervise, or recommend a contractor for the repair work.
- Corrections are verified on a site revisit to the same evidence standard, and the engineer closes each item individually.
- On a repairs required determination, the engineer also issues a sealed letter stating that the foundation did not comply, with the repair list. No certification is issued with an open repair item. There is no conditional certification.

11. Certification letter

- On a pass determination, the engineer issues a sealed certification addressed to the recipient recorded at intake.
- On a repairs required determination, the engineer issues a sealed letter to the same recipient stating that the foundation did not comply with the named requirement, with the repair list. The certification follows only after the corrections are verified on a site revisit.
- The certification identifies the property, the home by manufacturer, serial number, HUD label numbers, and date of manufacture, the date of inspection, the foundation and anchorage as observed, and every addition.
- The certification names the requirement it is written against and states that the foundation complies with it. It names no other requirement.
- The certification states that the inspection was visual and non-invasive, that it reflects condition on the date of inspection only, and that no representation is made about concealed or below-grade conditions.
- The certification does not address the home's own structure, and does not forecast future performance of the foundation.
- A copy of an issued certification may be sent again to the same recipient. It is not re-dated, re-addressed, or issued for another loan. A new inspection is required.
- The engineer does not certify a foundation that was not inspected under this protocol by this firm.

12. Exclusions

The following are declined or routed to another service line under its own engagement terms. The coordinator records the routing in the job file and informs the customer in writing.

- A home built before June 15, 1976. Declined.
- A home that is not yet installed, or is being moved. Declined until installation is complete.
- A site-built or modular home. Routed to the foundation certification line.
- An open insurance claim on the foundation, or active or threatened litigation involving it. Declined, with a referral to an independent firm.
- A lender requirement the engineer finds this protocol's evidence cannot support. Declined.

- A home in a flood zone with no flood elevation certificate, or with finished grade under the home below the 100-year flood elevation. Declined.
- No safe access under the home where the customer refuses to provide it. Declined.
- A customer request for a predetermined conclusion. This is an automatic decline and the request is logged verbatim.

13. Records and retention

- Each job file contains: the intake record and customer uploads, the engineer's acceptance record, the dispatch record, the completed Appendix B checklist, all photographs, the technician's notes, the engineer's review record and determination, any repair list and its closure records, and the issued certification.
- Job files are retained for ten years from the date of the certification. For a job that produces no certification, the ten years runs from the date of the engineer's last recorded determination or closure.
- Ten years is the firm's floor, reviewed against TBPELS rules.
- The compliance file holds the engineer's license records, technician qualification and training records, and every issued version of this protocol.

14. Protocol maintenance

- The engineer reviews this protocol annually, and within thirty days of any change to the governing requirement or to the firm's field application.
- Revisions are issued as a new version number with the effective date and the engineer's approval recorded in the document control table.
- This document follows the firm's house protocol format. The portal's four parts are carried in Appendices A through C. Appendix A carries both the intake questions and the customer uploads. Appendix D is the reference list.

Appendix A. Intake questions and customer uploads

Asked and collected by the coordinator before the job is created. A job is not dispatched until every required item is present.

PART 1: INTAKE QUESTIONS

Purpose & recipient

1. Who requires the certification? (lender name and loan type: FHA / VA / USDA / conventional / other)

Answer: _____

2. What does the lender's request say? (record exactly, or attach it)

Answer: _____

3. Who will the certification be addressed to?

Answer: _____

4. What deadline is the customer working against, if any?

Answer: _____

Property basics

5. Property address and county

Answer: _____

6. Manufacturer and serial number, if known

Answer: _____

7. Single-section or multi-section

Answer: _____

8. Year the home was installed at this site

Answer: _____

9. What the home sits on, as the customer understands it (piers and tie-downs / slab / perimeter wall / other)

Answer: _____

Flags: a YES on any of these routes to engineer review BEFORE dispatch

10. Was the home built before June 15, 1976?

Yes ☐ No ☐ If yes, describe: _____

11. Has the home been moved since it was first installed?

Yes ☐ No ☐ If yes, describe: _____

12. Are the HUD labels or the data plate missing?

Yes ☐ No ☐ If yes, describe: _____

13. Has anything been added to the home? (room, porch, deck, carport, roof-over)

Yes ☐ No ☐ If yes, describe: _____

14. Does the lender name a requirement other than the HUD Permanent Foundations Guide?

Yes ☐ No ☐ If yes, describe: _____

15. Is the property in a flood zone?

Yes ☐ No ☐ If yes, describe: _____

16. Has another engineer or inspector already declined to certify this foundation?

Yes ☐ No ☐ If yes, describe: _____

17. Is there an open insurance claim or litigation involving the foundation?

Yes ☐ No ☐ If yes, describe: _____

Access & safety (decides dispatch feasibility)

18. Is there an opening to get under the home, and is the space dry?

Answer: _____

19. Occupied or vacant? Accessible if occupied?

Answer: _____

Prior paperwork prompt

20. Do you have: the foundation plans the home was set on, an earlier foundation certification, installation report, tie-down or anchor paperwork, survey, flood elevation certificate, permits for additions?

Answer: _____

PART 2: CUSTOMER UPLOADS (collected at intake, before dispatch)

Required

- Photo of the front of the property (confirms address/structure)
- The lender's written request for the certification

Required if it exists

- Foundation plans or drawings used for the home's foundation
- Any earlier foundation certification or engineer letter

Collect when applicable

- Flood elevation certificate (required where the property is in a flood zone)
- Installation report or installer paperwork
- Anchor or tie-down system paperwork
- Permits and drawings for additions
- Appraisal pages noting additions or alterations

Appendix B. Technician on-site checklist

Completed on site by the technician. The checklist cannot be submitted incomplete.

PHOTOS

- Before starting, make a rough sketch of the home and any additions and mark North. Take a photo of it.
- Tick the item you are about to photograph, take a photo of the checklist, then take the photos for that item.
- Move on to the next item the same way.

JOB INFORMATION

Job / Dispatch No.: _____

Property address (full, incl. county):

Technician name: _____

Date / arrival time / departure time: _____

Weather during visit: _____

Contact on site (name, role, phone): _____

[] Photo of address marker (house number, street sign, or mailbox) [PHOTO]

HOME IDENTIFICATION

[] All four elevations of the home [PHOTO]

[] HUD label on each section, every digit readable [PHOTO]

[] Data plate inside the home (kitchen cabinet, bedroom closet, or electrical panel) [PHOTO]

[] Serial number stamped on the front cross member of the frame [PHOTO]

HUD label numbers: _____

Manufacturer / date of manufacture / serial number: _____

Wind zone and roof load zone from the data plate: _____

Number of sections ____ length (ft) ____ width of each section (ft) ____

Technician notes / readings:

SITE

☐ Ground slope and drainage on each side of the home [PHOTO]

☐ Standing water, ponding, or erosion at or under the home [PHOTO]

☐ Downspout and drain discharge points [PHOTO]

Technician notes / readings:

RUNNING GEAR

☐ Towing hitch: removed or still attached [PHOTO]

☐ Axles and wheels: removed or still attached [PHOTO]

Technician notes / readings:

PERIMETER ENCLOSURE

☐ Enclosure on each side: material and whether it is continuous [PHOTO]

☐ How the enclosure is fastened at the top and what it sits on at the bottom [PHOTO]

☐ Access opening, tape across width and height [PHOTO]

☐ Every vent [PHOTO]

Enclosure material (masonry / concrete / treated wood / vinyl / metal / none): _____

Access opening width x height (in.): _____ vents, count ____ typical size (in.) _____

Technician notes / readings:

UNDER THE HOME

☐ Overview from the access opening, and looking down the length of each chassis beam [PHOTO]

- ☐ Clearance from ground to the bottom of the chassis beam at the lowest point, tape in frame [PHOTO]
- ☐ Ground cover or vapor barrier, and any wet soil [PHOTO]
- ☐ Footings: typical and worst, tape across length and width, thickness where visible [PHOTO]
- ☐ Piers: typical and worst on each chassis beam: material, how stacked, cap, and shims [PHOTO]
- ☐ Tallest pier, tape in frame [PHOTO]
- ☐ Pier spacing along each chassis beam, tape in frame [PHOTO]
- ☐ Piers at the marriage line and under exterior door and large window openings (multi-section) [PHOTO]
- ☐ Any pier that is leaning, cracked, loose, or not in contact with the beam [PHOTO]
- ☐ How the chassis beam is connected to the pier or foundation (bolted / welded / clamped / resting only) [PHOTO]
- Footing size (in.): _____ pier material: _____ mortared or dry-stacked: _____
- Pier spacing (ft), each beam: _____ tallest pier (in.): _____ lowest clearance (in.): _____
- Piers counted ____ piers with a deficiency ____
- Technician notes / readings:

ANCHORAGE

- ☐ Every strap or tie on each long side: where it attaches to the home and where it attaches to the ground or foundation [PHOTO]
- ☐ What each strap is anchored to (screw-in ground anchor / concrete footing or slab / concrete deadman / other) [PHOTO]
- ☐ Anchorage at each end of the home, running lengthwise [PHOTO]
- ☐ Any strap that is loose, rusted, cut, or missing [PHOTO]
- ☐ Any engineered bracing system, with its label or tag [PHOTO]
- Straps each long side: ____ / ____ spacing (ft): _____ straps each end: ____ / ____
- Straps with a deficiency ____
- Technician notes / readings:

ADDITIONS

- ☐ Every addition: room, porch, deck, carport, awning, roof-over [PHOTO]
- ☐ Whether each addition stands on its own supports or hangs on the home [PHOTO]
- ☐ Footing and post under each addition [PHOTO]
- Technician notes / readings:

UTILITIES

[] Water, sewer, electric, and gas connections under and at the home [PHOTO]

Technician notes / readings:

CLOSE-OUT

[] Full perimeter walk: anything unusual [PHOTO]

[] Exception menu cleared for every non-applicable item

Total photos: _____

Technician signature / date: _____

Technician notes / readings:

Appendix C. Engineer decision rules

Applied by the engineer to each evidence package. One determination is recorded per review. The rules below state the PFGMH and Handbook 4000.1 requirements as the engineer applies them. Where the lender names another requirement, the engineer records the rules for that job in the job file.

PASS

- Package complete: every checklist item present or properly excepted
- Home identified by HUD label or data plate, built on or after June 15, 1976
- Floor area of at least 400 square feet
- Towing hitch, axles, and wheels removed
- Footings and piers sound, plumb, capped, and in contact with the chassis, whether mortared or dry-stacked
- Home attached to the foundation by anchorage the engineer finds carries uplift, sliding, and overturning in both directions
- Perimeter enclosure of masonry, concrete, treated wood, vinyl, or metal that is continuous, secured, and vented, with an access opening
- Ground under the home drains and is dry
- In a flood zone: finished grade under the home at or above the 100-year flood elevation shown on the flood elevation certificate

- Every addition stands on its own supports, or its load on the home and foundation is accounted for
- Utilities permanently connected

REVISE

- Any required photo missing, blurry, or missing its tape
- HUD label or data plate photo that cannot be read
- Piers or straps recorded from the access opening only
- Worst-condition coverage doubtful (all photos look selectively favorable)
- Counts or measurements inconsistent with photos

REPAIRS REQUIRED (certification withheld, sealed letter and repair list issued)

- Towing hitch, axles, or wheels still attached
- No anchorage, or anchorage the engineer finds does not carry the loads
- Anchorage missing in the lengthwise direction
- A strap that is loose, rusted through, cut, or missing
- A pier that is leaning, cracked, loose, or not in contact with the beam
- A pier with no footing, or a footing that is cracked or undermined
- No perimeter enclosure, or one that is broken, unsecured, or not continuous
- Standing water or active erosion under the home
- An addition hanging on the home with no support of its own
- Certification proceeds only after repairs are verified on revisit

SITE REVISIT (technician returns with an engineer-specified capture list)

- Part of the underside not reached on the first visit, return to complete it
- Footing size or anchor type not established, return after the owner exposes one footing or anchor and photograph it
- Conflicting evidence (owner story vs. photos vs. uploads), return for the specific items that settle it
- Repairs required, return after repairs to verify each item on the repair list

DECLINE

- Home built before June 15, 1976
- Home not yet installed, or being moved
- Site-built or modular home, route to foundation certification line
- Open insurance claim or litigation, decline, with a referral to an independent firm
- Lender requirement this protocol's evidence cannot support
- In a flood zone: no flood elevation certificate, or finished grade under the home below the 100-year flood elevation
- No safe access under the home and customer refuses to provide it
- Customer requests a predetermined conclusion, automatic decline, log verbatim request

Appendix D. References

Item	Location
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Permanent Foundations Guide for Manufactured Housing, HUD-7584 (Handbook 4930.3G), September 1996	huduser.gov
FHA Single Family Housing Policy Handbook 4000.1, Manufactured Housing	hud.gov
24 CFR 203.43f, Eligibility of Mortgages Covering Manufactured Homes	ecfr.gov
Manufactured Home Construction and Safety Standards, 24 CFR Part 3280	ecfr.gov
Model Manufactured Home Installation Standards, 24 CFR Part 3285	ecfr.gov
Texas Manufactured Housing Rules, 10 TAC Chapter 80	tdhca.texas.gov
Texas Engineering Practice Act, Texas Occupations Code Chapter 1001	statutes.capitol.texas.gov
TBPELS rules, 22 TAC Chapter 137	pels.texas.gov
254-RC-001 Roof Certification Protocol, v1.1, issued September 18, 2026	Firm document

Approval

Reviewed and approved by the Engineer of Record. This protocol is in force from the date below.

Aman Dhakal, P.E.

Engineer of Record, 254 Engineering Services

Signature: _____ Date: _____